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Assignment #01

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Question:01

(a)

Given that.

Client cache RTT = 20 ms

Cache access = 5 ms

so,

Total cache hit = 20+5 = 25 ms.

(b)

Given that:

Client cache = 20 ms

Cache access = 05 ms

Origin RTT = 100 ms

Total time = 20+5+100 = 125 ms.

(c)

Given that:

%age of cache hit = 80%

%age of cache miss = 100% - 80% = 20%

Time taken (cache hit) = 25ms

Time taken (cache miss) = 125ms.

Avg. = (0.8 x 25) + (0.2 x 125) = 45ms

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Question: 02

(a)

Given that:

Total requests = 10,000

Hit rate = 70%.

Hence, servings = 10,000 × 0.7 = 7,000 requests/hr.

(b)

Given that:

Total requests = 10,000

Cache served = 7,000

Hence, server served = 10,000 - 7,000

= 3,000. requests/hr.

(c)

Given that:

Hit rate = 85%. , Requests = 10×10^3

Total requests per hour = 0.85 × 10,000

= 8500

Hence, requests to server = 10,000 - 8,500

= 1,500.

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Yes, I will recommend this approach.

Reasons:

- Because it reduces the transmission & response time "x" since, the cache is relatively very close to the clients than the origin/main servers.

- Because it contributes in reducing the traffic on the main link/server.

Question : 04

(a)

Problem: In this case, the client will receive an outdated copy lacking the updated info.

(b)

Conditional GET: Conditional GET is sent to the sever & the server responds as:

- Code 200 OK, if object not changed
- Code 301 incase of updation of object.

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Structure:

if-modified since : <date>.

Question: 05

a) Persistent: HTML file & all 5 elements will be transported over the same single connection.

b) Non-persistent: In case of non-persistent connection, 6 connections will be required. One for HTML file & 5 for one of each element.

Question: 06

HTTP/2 supports objects multiplexing or concurrency meaning multiple objects can be sent over the same TCP connection simultaneously.

- Normal sequential transfer: Objects follow simple sequential approach. Object A(100 KB) is downloaded first & the D(20 KB) at the end.

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- Multiplexing (HTTP/2): Data from A, B, C, D can be interleaved over the same connection, allowing smaller objects to make progress while larger files (e.g A) are still downloading.

* Problem solved: One large or slow object delaying the transfer of the other objects.

Question #07

HTML & CSS will be/should have higher priority, since browser needs them to render the page. Hence, the usable page should be prioritized over the large image or video files.

Question #08

Yes, HTTP/3 will be more suitable for this user.

HTTP/2	HTTP/3
• Runs on TCP	• Runs on QUIC.

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• Uses IP address & ports.	• Uses the connection ID.
• Connection breaks when IP changes.	• Connection continues when IP changes.
• Possibility of the delay.	• No possibility of delay (no delay).

Question #09.

(a)

No, TCP delievers data in a strict order, so if packet is A is lost, B & C also have to wait.

(b)

HTTP/3 uses QUIC, in which each data stream is managed seprately. If A is lost, only A will be delayed & B and C will keep going.

Question # 10.

Caching strategy:

We should use a proxy at the campus edge. It will keep a copy of popular updates or data so

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that next users can request them from the local server (network) instead of requsting from the internet.

HTTP Strategy:

We should use the caching headers that tells the browser & the proxy to keep a file & use it again instead of ~~using it a~~ downloading it again & again.

fin!